

Autonomous Pet Waste Collection Robot



Team: Doggie Doo

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Problem Statement & Motivation

The Problem

Pet waste cleanup is an inconvenient, repetitive chore for millions of dog owners. Beyond inconvenience, uncollected waste spreads pathogens into lawns, soil, and waterways.

Market Gap

Existing tools — bags, scoops, and rakes — still require direct user involvement. No widely available autonomous consumer solution exists today.

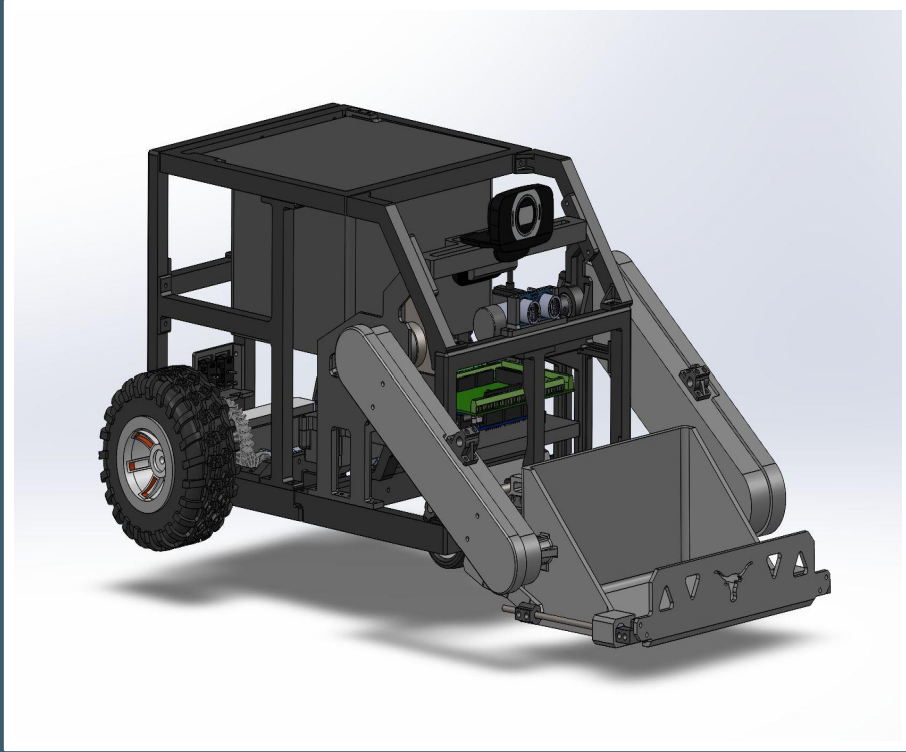
Our Objective

Design and validate a proof-of-concept autonomous robot that detects, collects, and stores pet waste with minimal user interaction in a residential backyard environment.

Engineering Specifications

ID	Category	Specification	Metric
1	Mobility	Minimum forward speed	0.5 ft/s
2	Mobility	Minimum backward speed	0.5 ft/s
3	Mobility	Minimum rotational speed	15 deg/s
4	Control	Speed tracking accuracy	±10%
5	Control	Bucket arm angular position accuracy	±5°
6	Functionality	Lateral alignment at pickup centerline	±2 in
7	Functionality	Forward stop accuracy at nominal pickup range	±0.5 in
8	Perception	Perception pipeline update rate	≥15 Hz
9	Perception	Pet waste detection range outdoors	1 – 10 ft
10	Safety	Stationary obstacle detection range	6 – 20 in
11	Safety	Minimum stopping clearance during forward motion	6 in
12	Functionality	Bucket payload lifting capacity	≥1.25 lbf

Isometric Views of CAD Assembly

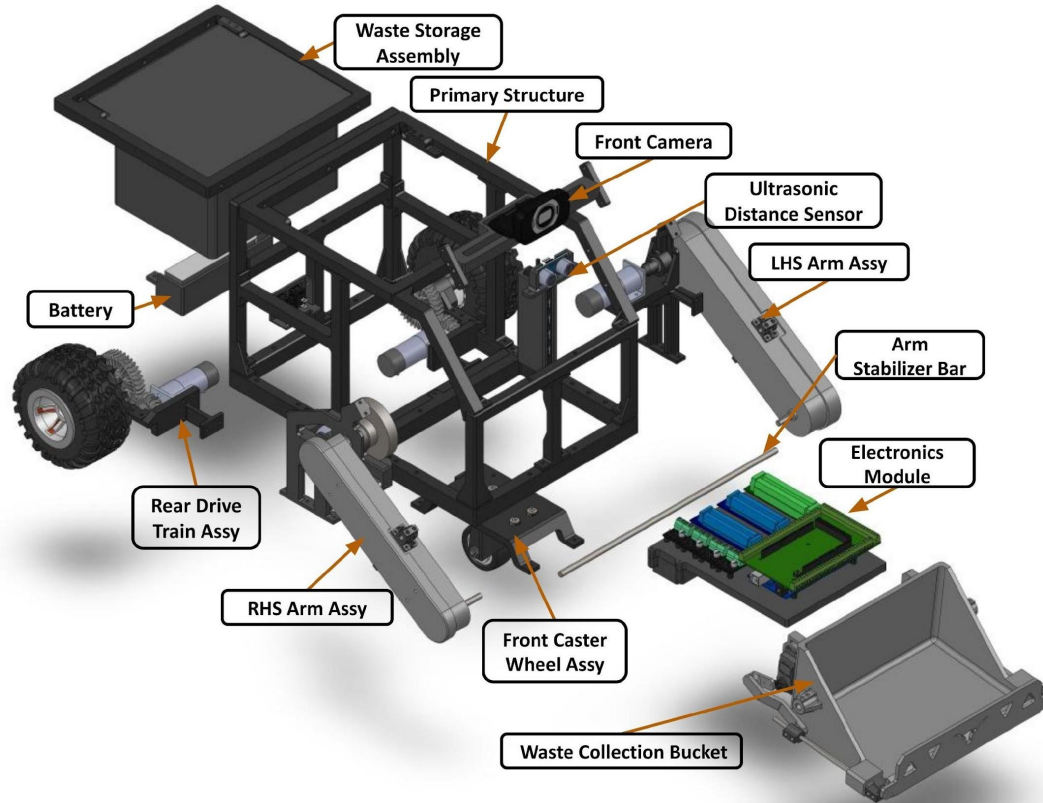


Collecting Position



Depositing Position

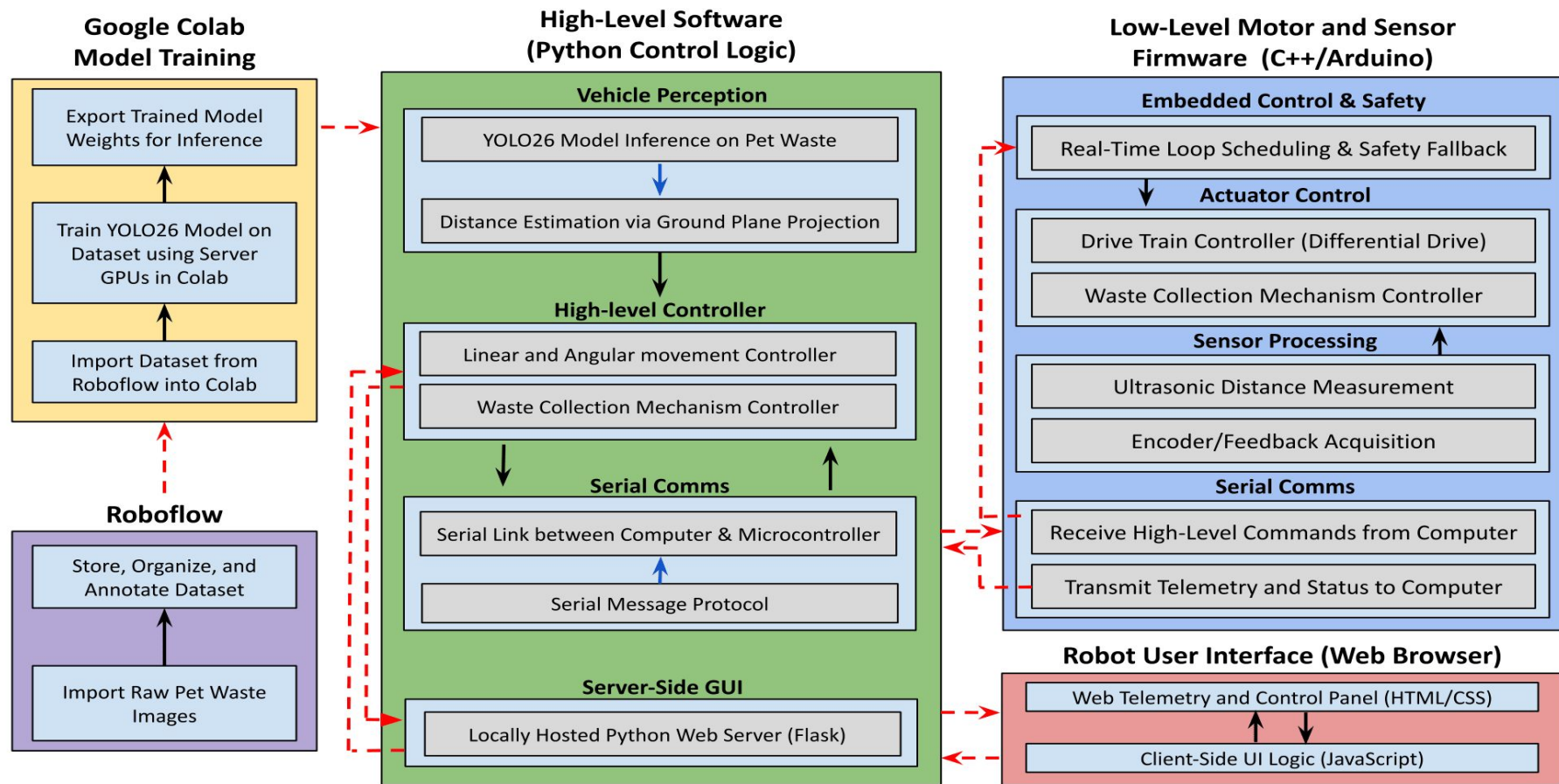
System Architecture — Exploded View



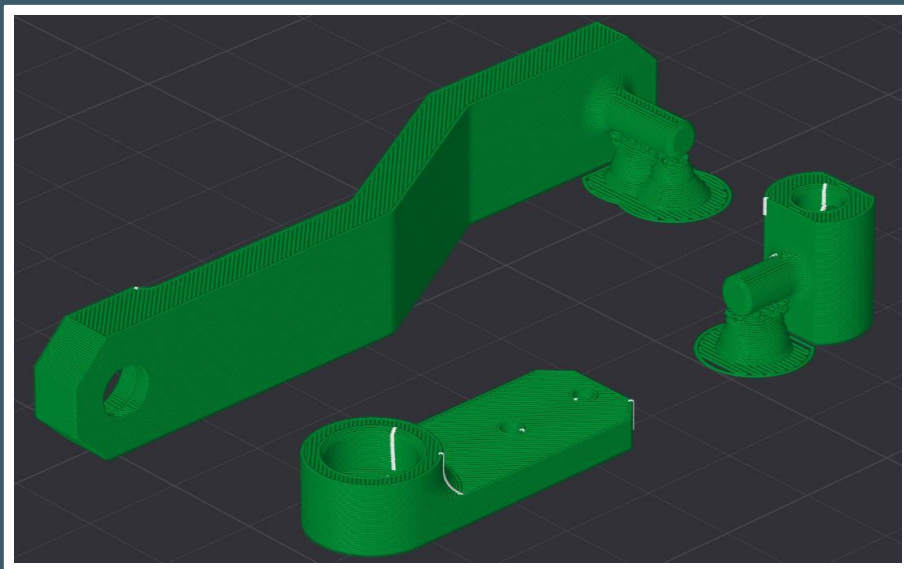
Key Subsystems

- Primary PLA frame structure
- Rear differential drivetrain
- RHS / LHS lift arm assembly
- Waste collection bucket + sweeper
- Front caster wheel assembly
- Front camera (YOLO detection)
- Ultrasonic distance sensor
- Electronics module (Arduino + drivers)
- Waste storage assembly
- 3S LiPo battery

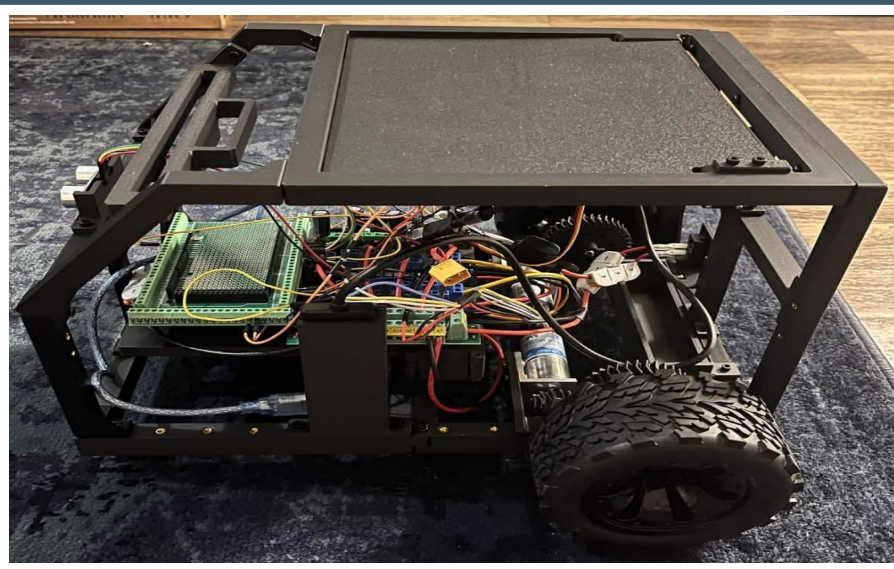
Software Architecture



Manufacturing — 3D Printing & Assembly



Sliced PLA linkage components (slider preview)



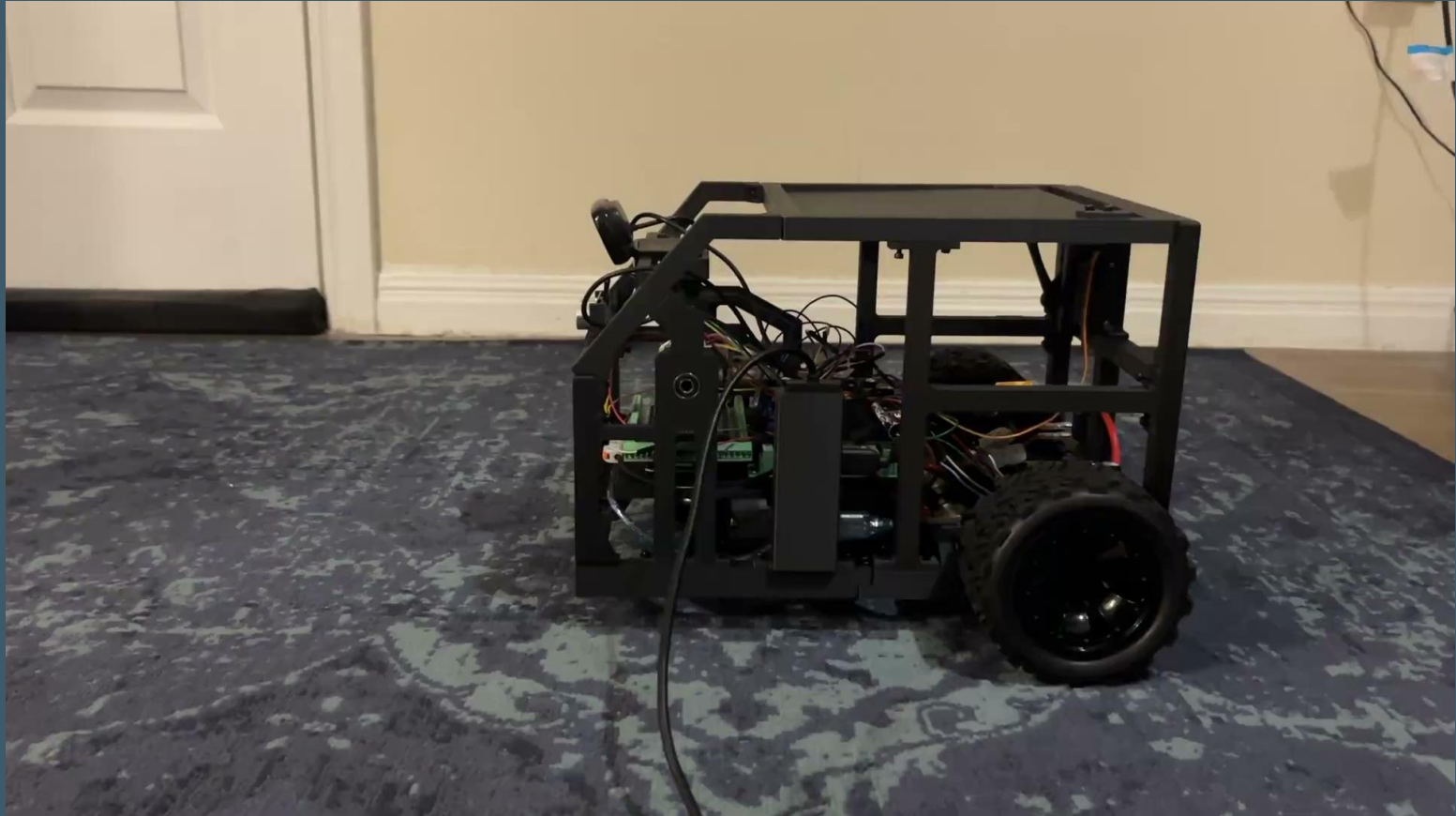
Fully assembled prototype — side view

- Chassis, drivetrain mounts, gears, linkages, brackets printed in PLA using FDM
- Steel rotary shafts cut on horizontal band saw — ensures bearing/gear fitment
- Rapid iteration: bucket Rev 1→2 replaced 3D-printed rod with steel linear shaft + nylon bushings

Collection Mechanism Movement



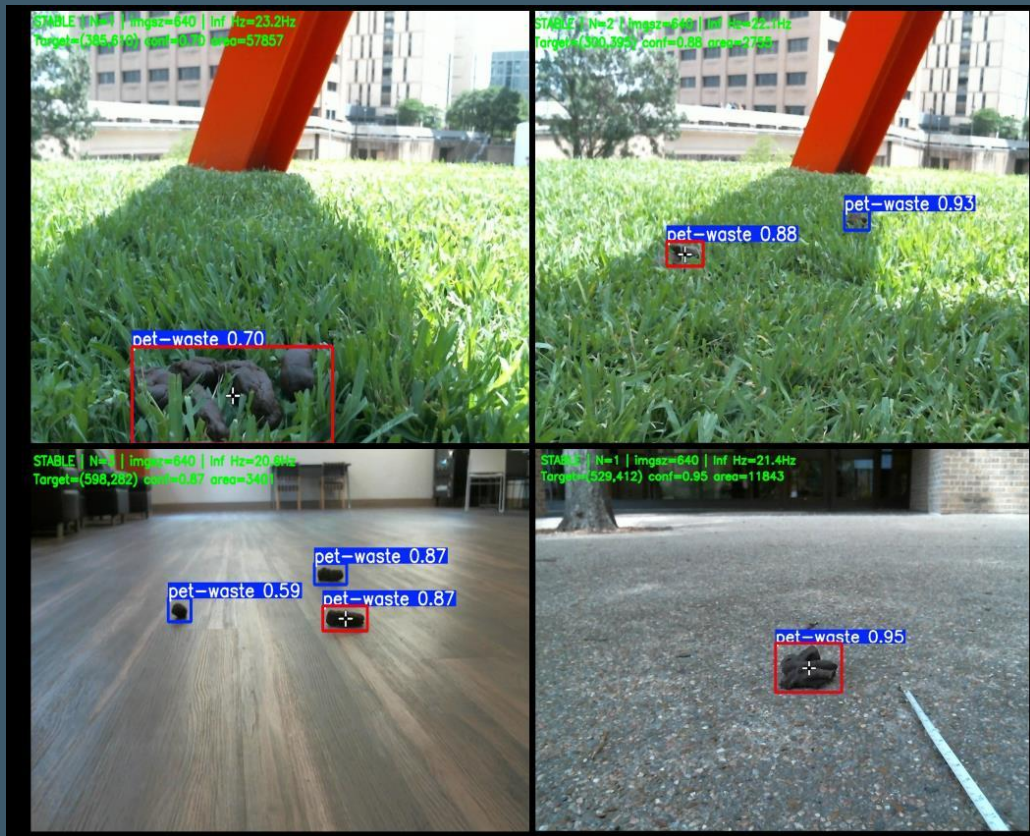
Forward and Backward Movement



Turning Movement



Pet Waste Detection — YOLO26 Nano Model



Detection on grass (outdoor) and hardwood (indoor)

Model Performance

0.86

Peak F1-Score

0.38

Optimal Confidence

15–25 Hz

Inference Rate

697

Training Images

Demonstration



Testing Results Summary

ID	Category	Specification	Status
1–3	Mobility	Forward/backward ≥ 0.5 ft/s; rotation ≥ 15 deg/s	PASS
4	Control	Speed tracking $\pm 10\%$ (avg meets; worst-case $\sim 14\text{--}16\%$)	PASS
5	Control	Bucket arm position $\pm 5^\circ$; measured max error 2.4°	PASS
6	Functionality	Lateral alignment ± 2 in from pickup centerline	PASS
7	Functionality	Forward stop ± 0.5 in of nominal pickup range	PASS
8	Perception	Pipeline ≥ 15 Hz; achieved 15 Hz (CPU) / 23–25 Hz (GPU)	PASS
9	Perception	Detection 1–10 ft outdoors; reliable to ~ 6 ft only	FAIL
10	Safety	Obstacle detection 6–20 in; error 0.7–1.6 in	PASS
11	Safety	Min stopping clearance 6 in; measured 5.8–6.8 in	PASS
12	Functionality	Bucket lift ≥ 1.25 lbf; achieved 1.75 lbf	PASS

Conclusions & Future Work

What We Demonstrated

- 11 / 12 engineering specifications verified
- Stable drivetrain and mechanism control
- YOLO26 at 0.86 F1-score, 15–25 Hz inference
- Bucket lifted 1.75 lbf — 40% above requirement
- Autonomous approach and alignment on flat surfaces
- Manual pickup sequence demonstrated successfully

Future Improvements

- Redesign drivetrain for grass: redistribute weight, add driven wheels or tracks
- Expand training dataset with more outdoor/grass images
- Integrate Raspberry Pi 5 onboard (remove laptop dependency)
- Replace failed gearbox with higher-quality actuator
- Add LiDAR for SLAM navigation and robust obstacle avoidance
- Fully autonomous end-to-end pickup + storage validation